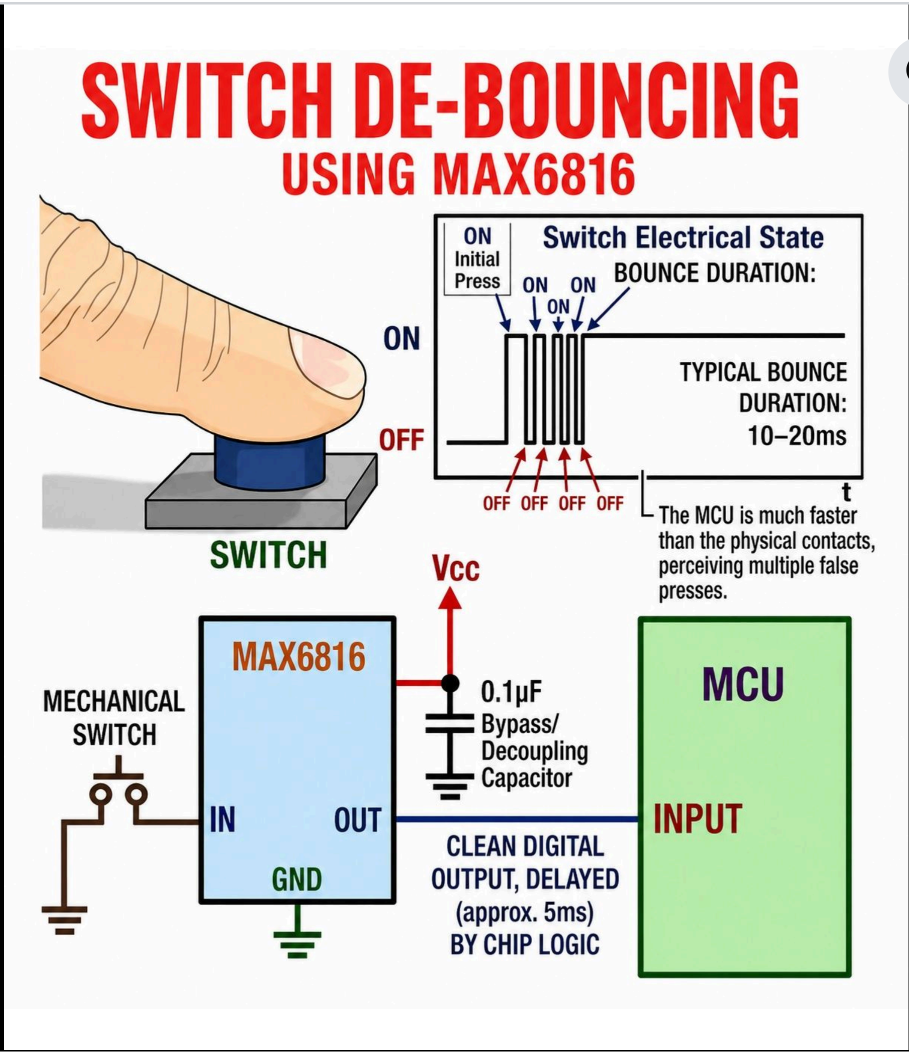

Electronics Education
5 maj kl. 04:50 ·

Mechanical switches don't change state cleanly. When you press them, the contacts physically bounce for a few milliseconds, creating multiple rapid ON/OFF pulses instead of one clean signal. A microcontroller reads these as multiple presses, which can cause glitches, false triggering, or incorrect counting.
The MAX6818 solves this by acting as a dedicated hardware debouncer. It takes the noisy switch input and outputs a clean, stable digital signal. Internally, it filters out the bounce and adds a controlled delay (typically around 20ms), ensuring only one valid transition is passed to the MCU. Key features include guaranteed debounce timing, no need for external RC components, and built-in pull-up resistors for simpler wiring. It also provides ESD protection and supports multiple channels, making it reliable for real-world button inputs. The result is consistent, noise-free switching behavior without software complexity, freeing up MCU resources and improving system reliability.
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2 v Gilla Svara 2

Michael Enggaard Pedersen
I would only use a debounce circuit in a pure logic circuit. With a CPU as the receiver, I would always debounce in software. 🙂

2 v Gilla Svara

Mgmgthant
You can use only a small capacitor parallel with the button.

2 v Gilla Svara

Darrell Wegner
And GM engineers still haven't figured out how to use that on their headlight switches.

2 v Gilla Svara

Stephen Glose
just program it out in the PLC program

Skriv en kommentar ...